

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

WE LOVE BIOLOGY

GET A CRASH COURSE IN THE LATEST DEVELOPMENTS in biological sciences along with things you've been missing out all along. Explore rapid evolution on our own planet, or head to Venus on India's very own terraforming mission. If that isn't personal enough for you, get down and dirty with DNA as we explore gene-editing and the concept of biological age.



Want daily eclipses? Head to Mars

Phobos and Deimos, both orbit very close to Mars. While Phobos rises and sets twice every Martian day of 24hr 40m, Deimos clocks a more leisurely 30 hours or so. Because of the proximity the Martian surface, a partial eclipse happens everyday. <http://dgit.in/LunarMars>

WHAT'S NEW

SpaceX and Boeing to launch Astronauts in 2018

In 2014, NASA contracted both Elon Musk's SpaceX and Boeing to develop space crafts that could ferry astronauts into space. SpaceX came up with the Dragon program (now Dragon 2) and Boeing unveiled the CST-100 Starliner program. Owing to a multitude of different reasons including budget cuts, both companies had to postpone their first flights to 2018.

▪ SpaceX Dragon 2

Planned to launch on an upgraded version of SpaceX's Falcon 9 rocket, known as Block 5, the Dragon 2 can ferry up to 7 crew members into Low Earth Orbit or to the ISS.



▪ Boeing CST-100 Starliner

Set to fly its first crew to space in June 2018, Boeing's Starliner also boasts of a 7-crew capacity like the Dragon 2.

<http://dgit.in/Spacer2018>

Converting digital DNA code to print viruses

Scientists have successfully built a prototype capable of transmitting genetic instructions using a digital medium and converting that data to create an actual organism. Dubbed as a biological teleporter, the machine from Scientists from

Synthetic Genomics Inc. (SGI) is capable of sending out a digital code of DNA to have them printed or manufactured at the

receiving end. The entire device is essentially a combination of smaller machines and robots. Still in prototype stage, the



"digital-to-biological converter" ensures that the conversion of the digital code is possible. After a series of experiments from

last year, they were successfully able to transmit genetic code digitally to the converter from another machine at a different location in the building.

As a trial, the DNA of the common flu virus was transmitted and automatically printed by the machine. This was the first time when a virus was automatically manufactured using DNA.

<http://dgit.in/reprintDNA>



Heating Moon Rocks

Scientists are exposing the lunar samples from the Apollo missions to heat. The rocks, known as regolith breccia 15498 which dates back 1 billion to 2.5 billion years were heated to over 1400 degrees in a lab chamber to study its magnetism.

<http://dgit.in/HeatMoon>



Patient care with machine learning

Two research teams from MIT's Computer Science and AI Labs (CASIL) have come up with new techniques employing Machine Learning in healthcare. The research utilises many different types of medical data, including electronic health records to predict outcomes in hospitals.

<http://dgit.in/MLhealth>



Human organs on animals

The idea of transplanting organs from one species into another has been around for a very long time now, but there has always been insurmountable problems that couldn't be solved. However, looking into these issues has become much easier thanks to CRISPR.

<http://dgit.in/HumanPig>